

The Effect of β -Blockade on Platelet Function and Fibrinolytic Activity

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Summary: Two groups of hypertensives and a group of migraine sufferers were tested during treatment with the nonselective β -blocker propranolol and the β_1 -selective metoprolol. During treatment with propranolol, an increased platelet aggregability and a decrease in platelet

content of cyclic AMP were seen when compared with metoprolol treatment. In addition, propranolol treatment increased the plasma level of adrenaline as well as the euglobulin clot lysis time. **Key Words:** β -Adrenoceptor blockade—Platelet function—Fibrinolysis.

During the past 10 years, it has been established that selective as well as nonselective β -blockers protect against myocardial infarction and angina pectoris, decrease the attack rate of migraine, and lower blood pressure (1-4).

It is also known that β -adrenoceptor blockade inhibits cyclic AMP formation in platelets, depending on their selectivity for β -adrenoceptors (5,6), and that the intracellular level of cyclic AMP modifies platelet aggregation (5,7). It has further been shown that the administration of drugs that increase cyclic AMP production results in enhanced fibrinolytic activity, which protects against thromboembolic complications (8). The purpose of this study was to investigate whether β -blockade interferes with different aspects of the coagulation system, especially focusing on platelet aggregability, platelet cyclic AMP content, plasma levels of cyclic AMP and adrenaline, and the fibrinolytic system.

PATIENTS

Three parallel studies were performed, each involving a group of 12 volunteers. *Group 1* consisted of patients suffering from mild hypertension (mean age, 62; range, 42-79 years) who were randomly allocated to either metoprolol 100 mg b.i.d. or propranolol 80 mg b.i.d. After two weeks of treatment, the group initially receiving metoprolol was changed to propranolol and vice versa. ADP-induced platelet aggregation and platelet cyclic AMP level were measured during each treatment period.

Group 2 included a similar group of hypertensive patients (mean age, 61; range, 48-74 years) who were randomly allocated to the same two regimens and crossed

over as previously described. Before entering the study and after each treatment period, fibrinolytic activity, defined as euglobulin clot lysis time (ECLT) before and after 5 min of venous compression, and plasma adrenaline and noradrenaline levels were measured.

Group 3 consisted of patients suffering from classic migraine (mean age, 40; range 30-49 years) who after initial blood sampling were allocated to either metoprolol 50 mg b.i.d. or propranolol 40 mg b.i.d. treatment. After 1 month, the medication was changed as previously described. Before and during each treatment period, platelet aggregation, plasma cyclic AMP, and platelet response to β -adrenoceptor stimulation in vitro (isoprenaline 10^{-5} M) were tested.

METHODS

Antecubital vein blood for determination of biological variables and routine hematology was drawn after 15 min in the supine position. Eating and smoking were not permitted during the 12 h before blood sampling, which invariably occurred at 9.00 a.m., 2 h after intake of the last tablet. During the study period the volunteers were not permitted to take any medicine but β -blockers, in a few cases combined with diuretics in a constant dosage.

Platelet aggregation. Citrated platelet-rich plasma and platelet-poor plasma were prepared as described elsewhere (9). Platelet aggregation was studied turbidometrically using a Payton dual channel aggregometer. Aggregation was induced by ADP in final concentrations of 0.25, 0.5, 1.0, 2.0, 4.0, 8.0, and 16.0 μ M. Threshold values for aggregation were determined and defined as the lowest concentration of ADP to cause irreversible aggregation with at least an 80% difference in light transmission between platelet-rich and platelet-poor plasma.

Platelet cyclic AMP measurement. Blood was drawn into 5 ml tubes containing 0.5 ml 3.13% citrate. Platelets

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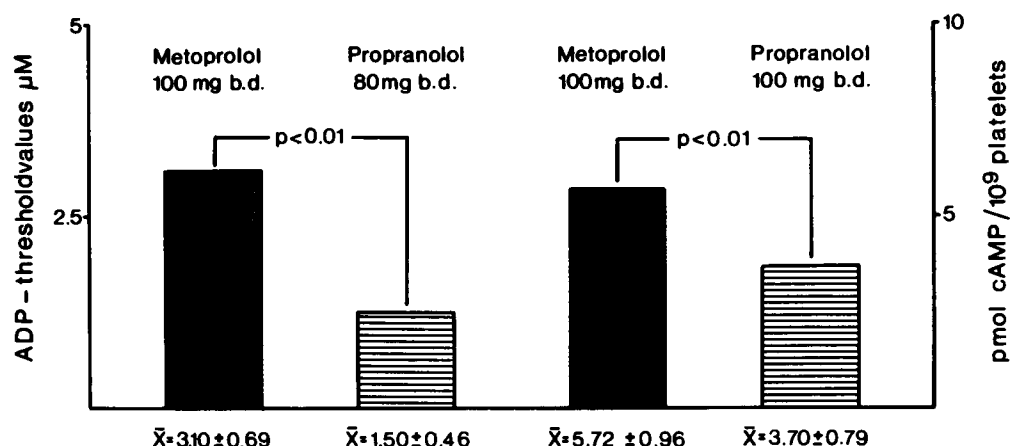


FIG. 1. ADP-threshold values producing irreversible platelet aggregation are shown on the left. Platelet content of cyclic AMP is shown on the right. Mean $\pm$ SEM, $n = 12$.

were isolated for determination of basal cyclic AMP level and cyclic AMP level after stimulation with isoprenaline (10^{-5} M), as described elsewhere, using a protein binding assay (6,10).

Plasma cyclic AMP measurements. Platelet-rich plasma was isolated (9) and centrifuged $2000 \times g$, 20 min, 4°C . 200 μl of the supernatant were isolated and stored at -60°C for later determination of cyclic AMP using the Becton Dickinson Radioimmunoassay.

Plasma catecholamines. Blood was collected in pre-cooled glass vials containing EDTA 90 mg/ml and glutathione 60 mg/ml and placed on ice for 20 min. After centrifugation $3000 g$, 20 min, 4°C , the supernatant was transferred to plastic vials and stored at -60°C for later determination using a double isotope technique (11).

Fibrinolysis. Fibrinolytic activity was determined as the euglobulin clot lysis time before and after 5 min of venous compression (12).

Statistics. For statistical evaluation, the Wilcoxon Signed-ranks matched-pairs test was used. $p \leq 0.05$ was regarded as a statistically significant difference. All values given are mean $\pm$ SEM.

RESULTS

Propranolol 80 mg b.i.d. resulted in a significant decrease in ADP thresholds to produce irreversible platelet aggregation when compared with metoprolol treatment (mean, 3.10 ± 0.69 ; and mean, 1.50 ± 0.46 μM , respectively; $p < 0.01$, Fig. 1). Correlating with the decrease in ADP thresholds, a significant decrease in platelet content of cyclic AMP was seen on comparison of propranolol and metoprolol treatment (mean, 5.72 ± 0.96 and 3.70 ± 0.79 pmol/10⁹ platelets, respectively; $p < 0.01$, Fig. 1).

Furthermore, during propranolol treatment a significant prolongation of the euglobulin clot lysis time (ECLT) was seen, as compared with metoprolol or no treatment (data not shown). This observation was even more pronounced on testing after 5 min of venous compression (mean, 483.64 ± 105.53 ; 274.60 ± 77.91 ; and 222.50 ± 60.58 min, respectively; $p < 0.05$, Fig. 2).

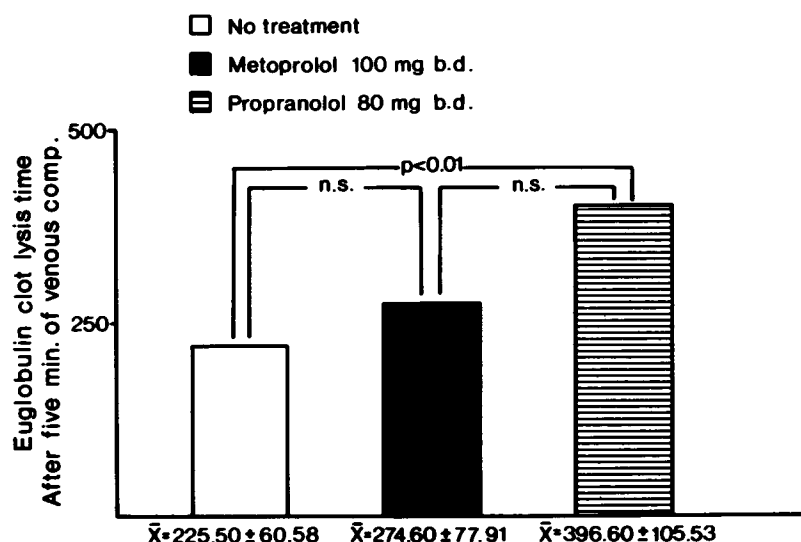


FIG. 2. Euglobulin clot lysis time before and during treatment with metoprolol and propranolol. Mean $\pm$ SEM, $n = 12$.

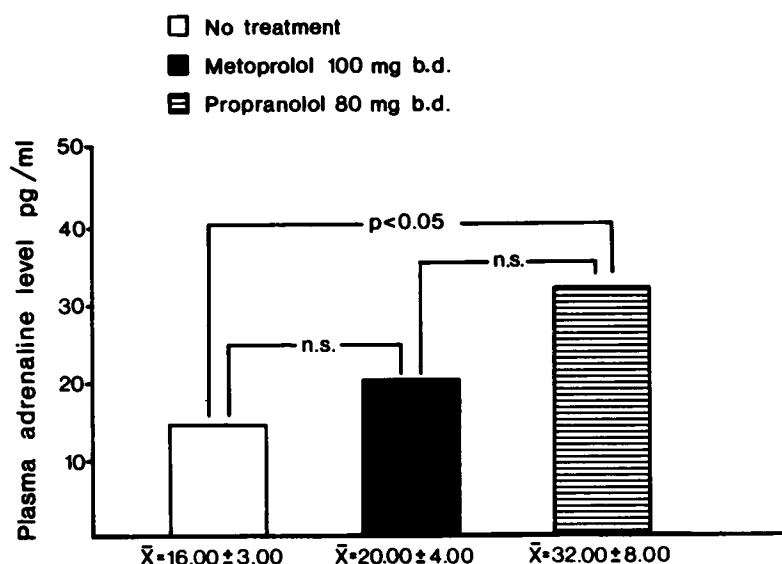


FIG. 3. Plasma adrenaline levels before and during treatment with metoprolol and propranolol. Mean $\pm$ SEM, $n = 12$.

The plasma adrenaline level increased significantly during propranolol therapy compared with no treatment (mean, 32.00 ± 8.00 and 16.00 ± 3.00 ng/ml, respectively; $p < 0.05$, Fig. 3). Metoprolol, however, did not cause any significant alteration in plasma adrenaline levels. No significant alteration in plasma noradrenaline levels was seen on comparison of the two different β -blockers (data not shown).

Propranolol 40 mg b.i.d. also caused a significant decrease in ADP thresholds to produce irreversible platelet aggregation, as well as a significant decrease in platelet cyclic AMP level when compared with metoprolol treatment (data not shown). In addition, propranolol 40 mg b.i.d. resulted in a significant decrease in plasma cyclic AMP level when compared with metoprolol treatment (mean,

1.50 ± 0.20 and 1.90 ± 0.30 pmol/100 μ l plasma, respectively; $p < 0.05$, Fig. 4).

Finally, propranolol 40 mg b.i.d. treatment resulted in a significant decrease in the platelet cyclic AMP content seen after isoprenaline stimulation in vitro when compared with the level seen after metoprolol 50 mg b.i.d. (30.10 ± 6.56 and 40.37 ± 6.18 , Fig. 5, $p < 0.05$).

DISCUSSION

These data indicate that treatment with the non-selective β -blocker propranolol results in an enhanced platelet aggregability, a decrease in platelet and plasma cyclic AMP level, a decrease in fibrinolytic activity, and β -adrenoceptor response to

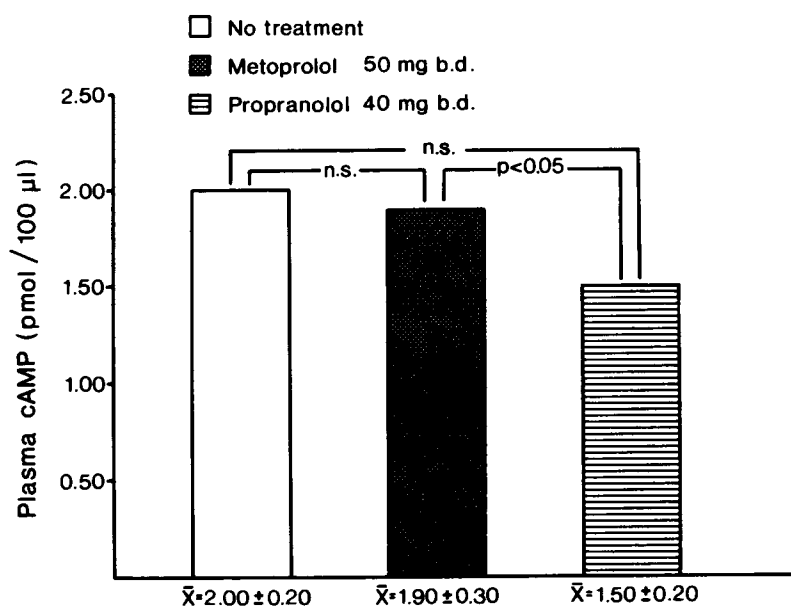


FIG. 4. Plasma levels of cyclic AMP before and after treatment with metoprolol and propranolol. Mean $\pm$ SEM, $n = 10$.

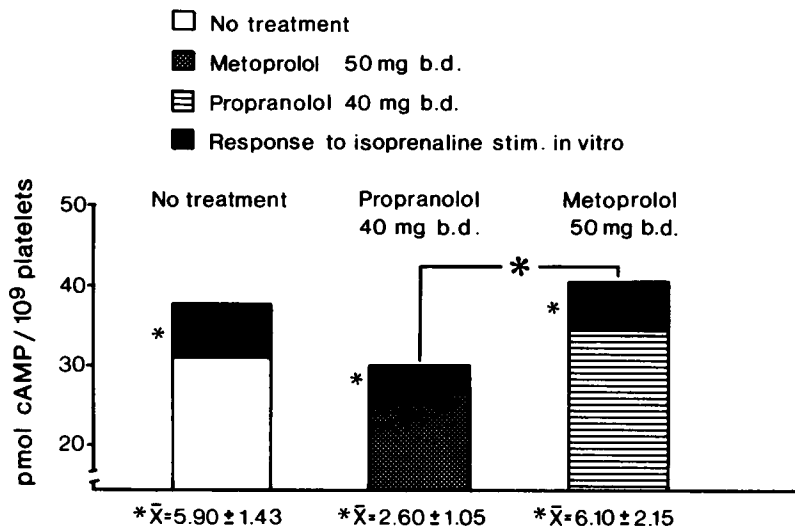


FIG. 5. Unstimulated and isoprenaline-stimulated platelet cyclic AMP values before and during treatment with metoprolol and propranolol. The response to isoprenaline is indicated as *mean $\pm$ SEM, $n = 12$.

isoprenaline stimulation in vitro, as well as an increased level of plasma adrenaline when compared with the β_1 -selective metoprolol.

It is known that platelets have α -adrenoceptors as well as β -adrenoceptors, and that β -adrenoceptor stimulation results in cyclic AMP production which inhibits platelet aggregation (13). The platelet β -adrenoceptor is mainly of the β_2 -subtype, and blockade of this receptor results in a lowering of the platelet cyclic AMP content as well as an enhanced platelet aggregability when testing with ADP and adrenaline (5-7). The present data are in agreement with these observations. During nonselective β -blockade, the circulating catecholamines, which stimulate α -adrenoceptors as well as β -adrenoceptors, act as "pure" α -agonists without the earlier described inhibitory effect of platelet β -adrenoceptor-induced cyclic AMP formation. The increased level of plasma adrenaline seen during nonselective blockade with propranolol might further enhance the aggregation by stimulating the platelet α -adrenoceptor.

In a similar study, the effect of the nonselective β -blocker timolol was studied. As with propranolol, timolol treatment resulted in enhanced platelet aggregability, a decrease in platelet cyclic AMP, and an increased level of plasma adrenaline, when compared with β_1 -selective or no treatment (14,15). In vitro stimulation of platelet β -adrenoceptors with isoprenaline resulted in a significant decrease in platelet cyclic AMP response during the propranolol regimen. This is possibly due to a blockade of platelet β_2 -adrenoceptors. The high lipophilicity of propranolol, however, might cause incorporation of the drug into the platelet membrane to such a degree that it cannot be washed out during preparation. In another study, treatment with timolol, which is a nonselective β -blocker with intermediate lipophilicity, no such

decrease in isoprenaline response occurred when compared with metoprolol (data not shown).

It has been reported that treatment with alprenolol, a nonselective β -antagonist with intrinsic sympathomimetic activity (ISA), results in prolongation of the ECLT (16). The present findings, which indicate a prolongation of the ECLT during treatment with propranolol, but not during treatment with β_1 -selective metoprolol, support these data and suggest that β_2 -adrenoceptor blockade decreases the fibrinolytic activity. The fibrinolytic process is regulated by a tissue plasminogen activator (t-PA). It has been shown that the addition of cyclic AMP enhances the production of t-PA in vitro (17). In our study, a decrease in plasma cyclic AMP was seen during propranolol treatment. A general lowering of plasma cyclic AMP might indicate a lowering of cyclic AMP production in cells responsible for t-PA production.

Finally, nonselective blockade increases the plasma level of triglycerides as well as cholesterol in comparison with β_1 -selective or no treatment (18). These alterations in lipid metabolism might produce a hypercoagulable state (19). In contrast to our data it has been reported that high doses of propranolol (640 mg/day) result in blockade of TxB_2 formation and inhibition of platelet aggregation (20). Using doses of different β -blockers of up to 200 mg/day, we have not been able to show any change in TxB_2 formation (data not shown).

In conclusion, moderate doses of nonselective β -blockade (propranolol) result in an enhanced platelet aggregability, as well as a decreased fibrinolytic activity when compared with the β_1 -selective metoprolol or no treatment. These alterations might be related to an altered plasma adrenaline level, a decreased production of intracellular mediators such as cyclic AMP, or to altered lipid metabolism.

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